# PULLBACK OF A PRINCIPAL DIVISOR ALONG AN ETALE MORPHISM IS THE SUM OVER THE IRREDUCIBLE COMPONENTS OF THE PUSHFORWARDS OF THE RESTRICTED DIVISORS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: a cycle-theoretic reduction to a coefficient comparison at generic points in henselizations, followed by a length computation over total rings of fractions that is finished by an additivity statement and a disjoint-union decomposition

*Editorial note.* The label lemma-etale-pullback-principal-divisor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Flat pullback of cycles with relative dimension 0, and the definition of the k-dimensional part of a cycle, so that the pullback of a divisor by an etale morphism is defined
- Deltas of points and their behaviour for etale morphisms, the generic point of an integral closed subspace and the scheme theoretic image construction for the induced closed subspace of the base
- Henselizations of reduced Noetherian local rings, their total rings of fractions, and the length formula for the order of a rational function along an integral closed subspace of codimension 1

*Source reference.* Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Prerequisite pointer invoked at proof line(s) 1777; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 02QG: sources/stacks-chow/tags.txt, source lines 3587–3587. Prerequisite pointer invoked at proof line(s) 1787; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original remark 0EPL: sources/stacks-chow/tags.txt, source lines 18381–18381. Prerequisite pointer invoked at proof line(s) 1726; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 02RB: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1262–1262. Prerequisite pointer invoked at proof

*Date*: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 1701; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EPY: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1332–1332. Prerequisite pointer invoked at proof line(s) 1742; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EEG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof line(s) 1715; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EE8: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 650–650. Prerequisite pointer invoked at proof line(s) 1689; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 1681; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 0EML: sources/stacks-chow/tags.txt, source lines 18311–18311. Prerequisite pointer invoked at proof line(s) 1697; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 02QL let $f : X \to Y$ be an étale morphism of good algebraic spaces over $B$. Assume $Y$ is integral. Let $g \in R(Y)^*$. As cycles on $X$ we have*

$$f^*(div_Y(g)) = \sum\nolimits_{X'} (X' \to X)_* div_{X'}(g \circ f|_{X'})$$

*where the sum is over the irreducible components of $X$ (Remark 0EE8).*

*Proof.* The map $|X| \to |Y|$ is open. The set of irreducible components of $|X|$ is locally finite in $|X|$. We conclude that $f|_{X'} : X' \to Y$ is dominant for every irreducible component $X' \subset X$. Thus $g \circ f|_{X'}$ is defined (Morphisms of Spaces, Section 0EML), hence $\mathrm{div}_{X'}(g \circ f|_{X'})$ is defined. Moreover, the sum is locally finite and we find that the right hand side indeed is a cycle on $X$. The left hand side is defined by Definition 02RB and the fact that an étale morphism is flat of relative dimension 0.

Since $f$ is étale we see that $\delta_X(x) = \delta_y(f(x))$ for all $x \in |X|$. Thus if $\dim_\delta(Y) = n$, then $\dim_\delta(X') = n$ for every irreducible component $X'$ of $X$ (since generic points of $X$ map to the generic point of $Y$, see above). Thus both left and right hand side are $(n-1)$-cycles.

Let $Z \subset X$ be an integral closed subspace with $\dim_\delta(Z) = n - 1$. To prove the equality, we need to show that the coefficients of $Z$ are the same. Let $Z' \subset Y$ be the integral closed subspace constructed in Lemma 0EEG. Then $\dim_\delta(Z') = n - 1$

too. Let $\xi \in |Z|$ be the generic point. Then $\xi' = f(\xi) \in |Z'|$ is the generic point. Consider the commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(\mathcal{O}^h_{X,\xi}) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathcal{O}^h_{Y,\xi'}) & \longrightarrow & Y \end{array}$$

of Decent Spaces, Remark 0EPL. We have to be slightly careful as the reduced Noetherian local rings $\mathcal{O}^h_{X,\xi}$ and $\mathcal{O}^h_{Y,\xi'}$ need not be domains. Thus we work with total rings of fractions $Q(-)$ rather than fraction fields. By definition, to get the coefficient of $Z'$ in $\mathrm{div}_Y(g)$ we write the image of $g$ in $Q(\mathcal{O}^h_{Y,\xi'})$ as $a/b$ with $a, b \in \mathcal{O}^h_{Y,\xi'}$ nonzerodivisors and we take

$$\mathrm{ord}_{Z'}(g) = \mathrm{length}_{\mathcal{O}^h_{Y,\xi'}}(\mathcal{O}^h_{Y,\xi'}/a\mathcal{O}^h_{Y,\xi'}) - \mathrm{length}_{\mathcal{O}^h_{Y,\xi'}}(\mathcal{O}^h_{Y,\xi'}/b\mathcal{O}^h_{Y,\xi'})$$

Observe that the coefficient of $Z$ in $f^*\mathrm{div}_Y(G)$ is the same integer, see Lemma 0EPY. Suppose that $\xi \in X'$. Then we can consider the maps

$$\mathcal{O}^h_{Y,\xi'} \to \mathcal{O}^h_{X,\xi} \to \mathcal{O}^h_{X',\xi}$$

The first arrow is flat and the second arrow is a surjective map of reduced local Noetherian rings of dimension 1. Therefore both these maps send nonzerodivisors to nonzerodivisors and we conclude the coefficient of $Z'$ in $\mathrm{div}_{X'}(g \circ f|_{X'})$ is

$$\mathrm{ord}_Z(g \circ f|_{X'}) = \mathrm{length}_{\mathcal{O}^h_{X',\xi}}(\mathcal{O}^h_{X',\xi}/a\mathcal{O}^h_{X',\xi}) - \mathrm{length}_{\mathcal{O}^h_{Y,\xi'}}(\mathcal{O}^h_{X',\xi}/b\mathcal{O}^h_{X',\xi})$$

by the same prescription as above. Thus it suffices to show

$$\mathrm{length}_{\mathcal{O}^h_{Y,\xi'}}(\mathcal{O}^h_{Y,\xi'}/a\mathcal{O}^h_{Y,\xi'}) = \sum\nolimits_{\xi \in |X'|} \mathrm{length}_{\mathcal{O}^h_{X',\xi}}(\mathcal{O}^h_{X',\xi}/a\mathcal{O}^h_{X',\xi})$$

First, since the ring map $\mathcal{O}^h_{Y,\xi'} \to \mathcal{O}^h_{X,\xi}$ is flat and unramified, we have

$$\mathrm{length}_{\mathcal{O}^h_{Y,\xi'}}(\mathcal{O}^h_{Y,\xi'}/a\mathcal{O}^h_{Y,\xi'}) = \mathrm{length}_{\mathcal{O}^h_{X,\xi}}(\mathcal{O}^h_{X,\xi}/a\mathcal{O}^h_{X,\xi})$$

by Algebra, Lemma 02M1. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ be the nonmaximal primes of $\mathcal{O}^h_{X,\xi}$ and set $R_j = \mathcal{O}^h_{X,\xi}/\mathfrak{q}_j$. For $X'$ as above, denote $J(X') \subset \{1, \ldots, t\}$ the set of indices such that $\mathfrak{q}_j$ corresponds to a point of $X'$, i.e., such that under the surjection $\mathcal{O}^h_{X,\xi} \to \mathcal{O}_{X',\xi}$ the prime $\mathfrak{q}_j$ corresponds to a prime of $\mathcal{O}_{X',\xi}$. By Chow Homology, Lemma 02QG we get

$$\mathrm{length}_{\mathcal{O}^h_{X,\xi}}(\mathcal{O}^h_{X,\xi}/a\mathcal{O}^h_{X,\xi}) = \sum\nolimits_j \mathrm{length}_{R_j}(R_j/aR_j)$$

and

$$\mathrm{length}_{\mathcal{O}^h_{X',\xi}}(\mathcal{O}^h_{X',\xi}/a\mathcal{O}^h_{X',\xi}) = \sum\nolimits_{j \in J(X')} \mathrm{length}_{R_j}(R_j/aR_j)$$

Thus the result of the lemma holds because $\{1, \ldots, t\}$ is the disjoint union of the sets $J(X')$: each point of codimension 0 on $X$ lies on a unique $X'$. $\square$